

Weighted recursive constructions for shattering triples with six permutations

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Abstract

Let $F_3(n)$ be the maximum number of triples of an n -element set that can be shattered by six permutations, and let $c_3 = \lim_{n \rightarrow \infty} F_3(n)/\binom{n}{3}$. Černá, Kielak and Volec recently proved

$$\frac{482}{975} \leq c_3 \leq \frac{1}{2}.$$

We give a weighted recursive construction that applies to every finite six-permutation template. If G is the set of triples shattered by a template on $[N]$, and $w_1, \dots, w_N$ is a probability vector with at least two positive entries, put

$$A = 6 \sum_{\{i,j,k\} \in G} w_i w_j w_k, \quad B = \sum_{i=1}^N w_i^3.$$

We prove that $c_3 \geq A/(1 - B)$. Uniform weights recover the standard iterated blow-up bound. Applying the weighted construction to the published 26-point template of Černá, Kielak and Volec gives

$$c_3 \geq \frac{1288385}{2599242} = 0.495677201276 \dots$$

The proof uses independent infinite words to define six lexicographic orders; conditioning on their first letters gives the renewal identity $D = A + BD$. All arithmetic for the 26-point application is verified by an accompanying exact certificate.

1 Introduction

A family of permutations of a finite set *shatters* a subset when every relative ordering of that subset occurs in the family. This is a permutation analogue of classical shattering in extremal set theory. Johnson and Wickes introduced the problem of maximizing the number of k -sets shattered by a fixed number of permutations [1]. In the first nontrivial complete case, six permutations attempt to realize all $3! = 6$ orders on as many triples as possible.

Write $F_3(n)$ for the maximum number of triples of $[n]$ shattered by six permutations, and put

$$f_3(n) = \frac{F_3(n)}{\binom{n}{3}}.$$

Johnson and Wickes proved that the limit

$$c_3 = \lim_{n \rightarrow \infty} f_3(n)$$

exists and developed a finite-template recursion. Černá, Kielak and Volec subsequently used flag algebras for the upper bound and a 26-point recursive construction for the lower bound

$$\frac{482}{975} \leq c_3 \leq \frac{1}{2}$$

[2]. Their lower construction iterates the finite template uniformly.

We allow the template classes to have unequal asymptotic masses. The main result is a weighted recursion theorem for an arbitrary finite template. Its proof uses independent infinite words over the template alphabet. Each of the six template permutations defines a lexicographic order on the words. For three sampled words, either their first letters form a shattered template triple, all three first letters coincide and the problem restarts on the tails, or the triple cannot be shattered. This gives an exact renewal equation.

For the 26-point template in [2], a rational weight vector improves the published uniform bound from $482/975 = 0.494358974359\dots$ to $1288385/2599242 = 0.495677201276\dots$. The general weighted theorem is the structural part of the result; the numerical bound is an exact application to the strongest publicly available finite template that we could verify.

The new lower bound remains below $1/2$, and no equality or matching upper bound is claimed. The authors of [2] continue to work on this problem, so we compare only with their accessible published construction.

2 Templates and weighted lexicographic recursion

Let

$$\Pi = (\pi_1, \dots, \pi_6)$$

be six rank-map permutations of $[N]$. We write $a <_{\pi_r} b$ if and only if $\pi_r(a) < \pi_r(b)$. A triple $T = \{i, j, k\} \subset [N]$ is *shattered by* Π if its restrictions in the six induced orders realize all six linear orders of T . Denote the set of shattered triples by $\mathcal{G}(\Pi)$.

Let $w = (w_1, \dots, w_N)$ satisfy

$$w_i \geq 0, \quad \sum_{i=1}^N w_i = 1,$$

and suppose at least two w_i are positive. Define

$$A(\Pi, w) = 6 \sum_{\{i,j,k\} \in \mathcal{G}(\Pi)} w_i w_j w_k, \quad B(w) = \sum_{i=1}^N w_i^3.$$

The first quantity is the probability that three independent w -letters are pairwise distinct and form a shattered template triple. The second is the probability that all three letters coincide.

Theorem 2.1 (Weighted recursion). *For every $n \geq 3$, there exist six permutations of $[n]$ that shatter at least*

$$\frac{A(\Pi, w)}{1 - B(w)} \binom{n}{3}$$

triples. Consequently,

$$\boxed{c_3 \geq \frac{A(\Pi, w)}{1 - B(w)}}.$$

Proof. Let $X_1, \dots, X_n$ be independent infinite words over $[N]$, with every letter in every coordinate independently distributed according to w . For each $r \in [6]$, order the alphabet $[N]$ by π_r , and order the words lexicographically using this alphabet order.

For two independent words, the probability that their first m letters agree is

$$\left(\sum_{i=1}^N w_i^2 \right)^m.$$

At least two weights are positive, so $\sum_i w_i^2 < 1$. Hence two infinite words coincide with probability zero. Since only finitely many words are sampled, almost surely every lexicographic order is a permutation of the sample indices.

Fix three sample indices, and let D be the probability that their triple is shattered by the six lexicographic orders. Classify the three first letters. If they are pairwise distinct and their unordered set lies in $\mathcal{G}(\Pi)$, the template realizes all six relative orders. This event has probability $A(\Pi, w)$.

If all three first letters are equal, their relative orders are determined by the independent tails. Conditional on this event, the probability of being shattered is again D . The event has probability $B(w)$.

If exactly two first letters agree, the word with the different first letter lies on the same side of the other two whenever the relative order of the two alphabet letters is fixed. Thus all six orders cannot occur. If the first letters are pairwise distinct but their template triple is not shattered, later coordinates cannot change their relative order, so the sampled triple again fails.

The cases are exhaustive, and therefore

$$D = A(\Pi, w) + B(w)D.$$

Since at least two weights are positive, $B(w) < 1$, and hence

$$D = \frac{A(\Pi, w)}{1 - B(w)}.$$

Every triple of sample indices has the same probability D of being shattered. The expected number of shattered triples is therefore $D \binom{n}{3}$, so at least one finite sample attains at least this value. If $D > 0$, any six orders shattering a triple are necessarily distinct. If $D = 0$, choose any six distinct permutations, which is possible for $n \geq 3$. This proves the finite statement, and taking the limiting maximum gives the asserted bound on c_3 . $\square$

3 Uniform recursion as a special case

Suppose $w_i = 1/N$ for every i , and let $s = |\mathcal{G}(\Pi)|$. Then

$$A = \frac{6s}{N^3}, \quad B = \frac{1}{N^2},$$

and Theorem 2.1 becomes

$$c_3 \geq \frac{6s}{N(N^2 - 1)}.$$

This is the standard uniform finite-template bound used in [1, 2]. Thus the weighted theorem extends the known recursion rather than replacing it with a different limiting object.

For a fixed template, the theorem gives the finite-dimensional optimization problem

$$\Phi_{\Pi}(w) = \frac{6 \sum_{\{i,j,k\} \in \mathcal{G}(\Pi)} w_i w_j w_k}{1 - \sum_i w_i^3}$$

over the probability simplex. Numerical optimization can suggest a weight vector, but every claimed lower bound can be certified with rational weights and exact arithmetic.

4 The 26-point certificate

Černá, Kielak and Volec give six permutations of [26] that shatter 1446 of the 2600 triples [2, Table 2]. Uniform recursion yields

$$\frac{6 \cdot 1446}{26(26^2 - 1)} = \frac{482}{975}.$$

For the same template, take $w_i = q_i/250$, where

$$(q_1, \dots, q_{26}) = (12, 7, 7, 8, 9, 14, 9, 9, 9, 16, 16, 9, 12, \\ 10, 6, 6, 8, 7, 7, 12, 8, 14, 8, 9, 9, 9).$$

The entries are positive integers with sum 250. Exact summation over the 1446 shattered triples gives

$$A = \frac{773031}{1562500}, \quad B = \frac{7387}{3906250}.$$

Corollary 4.1. *The limiting proportion of triples shattered by six permutations satisfies*

$$c_3 \geq \frac{1288385}{2599242} = 0.495677201276 \dots$$

The improvement over the published uniform construction is exact:

$$\frac{1288385}{2599242} - \frac{482}{975} = \frac{1113577}{844753650} = 0.001318226917 \dots > 0.$$

The accompanying checker stores the six rank maps printed in [2], enumerates all $\binom{26}{3} = 2600$ triples, confirms the count 1446, and evaluates A , B , the weighted density and the difference above using rational arithmetic. It also converts the rank-map convention to equivalent order lists and checks that both representations give the same shattered triples. This conversion is an implementation cross-check, not an independent source for the template.

5 Scope and reproducibility

The result supplies a general weighted construction and a strict improvement of the lower bound obtained from the accessible published 26-point template. It does not prove $c_3 = 1/2$, classify optimal templates, or improve the flag algebra upper bound in [2]. Continuous optimization was used only to locate the displayed integer vector; the theorem and final numerical comparison are exact.

Our literature check covered the journal and arXiv versions of [1], the EUROCOMB'25 extended abstract [2], the authors' public publication pages, and direct searches for weighted blow-ups and the exact constant. The accessible texts use uniform recursion and do not state Theorem 2.1 or Corollary 4.1. This records the checked boundary; it is not a claim that unpublished or unindexed parallel work does not exist.

The verification script is supplementary evidence for the finite template and arithmetic. The proof of Theorem 2.1 does not depend on the script.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

References

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